

Exercises

H Chan

Exercise 1.

A topological space is said to be *Hausdorff* if for any two distinct points in the space, there exists two disjoint open sets containing each point. Every metric space is automatically a Hausdorff space (see if you can prove this for yourself), but plenty non-Hausdorff spaces exist.

(a) (Difficulty 2/10)

Show that if a finite group G acts on a Hausdorff space Y such that no nontrivial element has a fixed point, the action is even. Recall that this means that for all $y \in Y$, there exists an open set U containing y such that the open sets $\{gU\}_{g \in G}$ are *pairwise disjoint*, i.e. $g_1U \cap g_2U = \emptyset$ for all $g_1, g_2 \in G$. By the result in lecture, this implies that if Y is connected then $Y \rightarrow G \backslash Y$ is a Galois cover.

If you are not comfortable with working with topological spaces, you may assume that Y is a metric space instead.

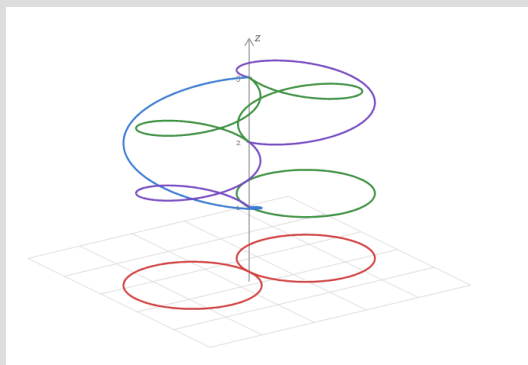
(b) (Difficulty 4/10)

Show that the finiteness of G is a necessary requirement, by considering the example where $Y = S^1 = \{(\cos(t), \sin(t)) \in \mathbb{R}^2 \mid t \in [0, 2\pi)\}$ is the unit circle and $G = \mathbb{Z}$; where $n \in \mathbb{Z}$ acts on Y via rotation by n radians counterclockwise; i.e. $n \cdot (\sin(t), \cos(t)) = (\sin(n+t), \cos(n+t))$. You need not verify that this is a well-defined group action, and you can take for granted the group also acts by *isometries*; i.e. $d(p, q) = d(n \cdot p, n \cdot q)$ for all $p, q \in S^1$ and $n \in \mathbb{Z}$.

Exercise 2.

(Difficulty 4/10)

Show that the following cover of the *figure 8* (denoted $S_1 \vee S_1$) is *not* Galois:



Exercise 3.

(Difficulty 6/10)

Show that a connected cover $Y \rightarrow X$ is Galois with automorphism group G if and only if the fibre product $Y \times_X Y$ is isomorphic to the trivial cover $Y \times G \rightarrow Y$; where G is assumed to carry the discrete topology.

Exercise 4.

(Difficulty 10/10)

Let $S^1 = \{(\cos(t), \sin(t)) \in \mathbb{R}^2 \mid t \in [0, 2\pi)\}$ denote the unit circle.

Show that every cover $p : S^1 \rightarrow S^1$ is Galois, and find its Galois group.

(If you are not comfortable with working with topological spaces, you may assume that every cover is already a metric space.)

(**Hint.** There are many ways to approach this problem. I outline one of them below, but you are welcome to try and figure out your own method.)

(1) Suppose that $p : S^1 \rightarrow S^1$ is some degree n cover. Construct a parameterization $f : [0, 2n\pi) \rightarrow S^1$ such that $(p \circ f)(t) = (\cos(t), \sin(t))$.

(2) Show that $f(t) \mapsto f(t + 2\pi)$ induces a covering automorphism of S^1 over S^1 .

(3) Show that there are exactly n distinct automorphisms, and how this implies that the cover is Galois. What is the group structure?)

If you have only listened to my first talk, do *not* attempt the questions after this page (you're welcome to take a look - they will just be nigh impossible to do without the theorems I will introduce in the second talk).

The second half of this document is meant to be done only after the second talk.

Warning: I have tried my best to keep the questions at a manageable difficulty level for those with a minimal mathematical background. Despite my best efforts, the rest of this document represents a *significant* difficulty jump in the exercises. You are regardless welcome to attempt them and ask me questions if you need help.

Exercise 5.

(a) (Difficulty 14/10)

Let X be a connected and locally path-connected, but not necessarily locally simply connected topological space. Show that every finite cover of X is *dominated* by some Galois cover; i.e. for every finite cover $\pi : Y \rightarrow X$ there exists a cover $Z \rightarrow Y \rightarrow X$ such that $Z \rightarrow X$ is Galois.

(**Hint.** Prove the question statement via the following steps:

(1) First argue why you can WLOG assume that Y is connected.

(2) Consider the n -fold fibered product of Y over X , denoted Y_X^n . Notice that this space admits a natural action of S_n .

(3) Let $\hat{Y}$ denote the connected component of Y_X^n that contains the point $(y_1, \dots, y_n)$, where $\{y_i\} = \pi^{-1}(x)$ is the fiber above some chosen basepoint x . This space still admits the same action of S_n .

(4) Notice that $\pi_1(X, x)$ embeds as a transitive subgroup of S_n via the monodromy action, and S_n has a well-defined morphism into $\text{Aut}(\hat{Y}|X)$. Argue why this implies that $\hat{Y}$ is Galois.)

(b) (Difficulty 14/10)

Show that the Galois cover you have constructed is *terminal* among all the Galois covers of Y over X : Namely, if $W \rightarrow Y$ is any Galois cover, then there exists a unique morphism $W \rightarrow Z$ such that the following diagram commutes:

$$\begin{array}{ccccc} W & & & & \\ \downarrow & \searrow & & & \\ Z & \longrightarrow & Y & \xrightarrow{\pi} & X \end{array}$$

(this is an instance of a *universal property*.)

Hence, we see that every finite cover of a space admits a *Galois closure*, which corresponds to the similar notion that we have for finite separable field extensions.

Exercise 6.

A *topological group* is a group G that is also a topological space, such that the group structure is compatible with the topology in the following sense: 1) group multiplication is a continuous function $G \times G \rightarrow G$ (where $G \times G$ is viewed with the product topology), and 2) inversion is a continuous function $G \rightarrow G$.

Examples of topological groups include $\mathbb{R}^n$ under addition, or the circle group S^1 under multiplication.

Let G be a connected and locally simply connected topological group with unit element e . Let $\pi : \tilde{G}_e \rightarrow G$ be a universal cover, and $\tilde{e}$ the universal element in the fiber $\pi^{-1}(e)$.

(a) (Difficulty 18/10)

Equip $\tilde{G}_e$ with a natural structure of a topological group with unit element $\tilde{e}$ for which π becomes a homomorphism of topological groups.

(b) (Difficulty 16/10)

Show that a group structure with the above properties is unique.

(c) (Difficulty 16/10)

Construct an exact sequence of topological groups

$$1 \rightarrow \pi_1(G, e)^{\text{op}} \rightarrow \tilde{G}_e \rightarrow G \rightarrow 1,$$

the topology on $\pi_1(G, e)$ being discrete.